

EGVLR: Evidence-Grounded Vision–Language Reinforcement for Anomaly Reasoning

Shih-Chih Lin¹, Ying-Heng Lu¹, Dong You Ye¹, and Shang-Hong Lai¹

National Tsing Hua University, Hsinchu, Taiwan
leolin65@gapp.nthu.edu.tw, s113062558@m113.nthu.edu.tw
114087510@office365.nthu.edu.tw, lai@cs.nthu.edu.tw

Abstract. Multimodal large language models have recently shown promise for industrial anomaly understanding, where a model is expected not only to decide whether an object is defective, but also to localize the supporting visual evidence and explain its decision. However, existing models often rely on language priors or shortcut reasoning when facing fine-grained industrial defects. This leads to inconsistent behaviors, such as correct answers with invalid locations, hallucinated defect boxes on normal samples, or rationales that contradict the final prediction.

We propose **EGVLR**, an evidence-grounded vision–language reinforcement framework for industrial anomaly reasoning. EGVLR enforces a unified diagnostic response format that explicitly separates visual evidence, diagnostic logic, spatial location, and the final answer. The framework progressively teaches the model to attend to localized defect evidence, incorporate industrial-domain reasoning, and optimize answer–location–rationale consistency through geometry- and semantics-aware preference optimization. For visualization, predicted boxes can be further converted into dense masks using an off-the-shelf segmentation backend. Experiments on MMAD show that EGVLR improves evidence-grounded anomaly reasoning, with consistent gains in defect localization, reasoning-dependent question answering, and spatial false-positive control. These results suggest that explicitly coupling answers, locations, and rationales is important for reliable industrial anomaly understanding. Project page: <https://github.com/leolin65/EGVLR-project>.

Keywords: Industrial Anomaly Reasoning · Multimodal Large Language Models · Visual Grounding · Reinforcement Learning

1 Introduction

Industrial anomaly inspection is moving from closed-set defect scoring toward interactive anomaly understanding. In a conventional pipeline, an anomaly detector returns an image-level score or a pixel-level heatmap; a human inspector must still decide what the defect is, whether the highlighted region is meaningful, and how the query differs from a normal product. Prior industrial anomaly detection systems have improved representation learning, synthetic defect modeling, quantized reconstruction, and efficient multi-class deployment [15, 18–23, 29],

but they still primarily output scores, heatmaps, or masks rather than diagnostic language. Recent multimodal large language models (MLLMs) make a richer interface possible: they can receive inspection images and natural-language questions, compare a query with a normal reference, describe visible deviations, and return structured answers [5, 11, 14, 24, 31, 32]. Benchmarks and resources such as MMAD [9] therefore evaluate not only whether a model detects an anomaly, but also whether it can reason about defect category, location, description, and analysis.

This setting exposes a gap between general visual-language capability and industrial inspection reliability. Industrial defects are usually sparse, local, and visually subtle. A small crack, contaminant, missing component, solder defect, or abnormal texture may occupy only a tiny region, while the rest of the image remains visually normal. Moreover, the meaning of a deviation is category-dependent: a texture change that is harmless for one product may be a defect for another. General MLLMs are mostly trained on broad web image-text data, where such localized industrial evidence is rare and where answers are not tied to precise spatial evidence. As a result, they may answer inspection questions from language priors instead of image evidence. A model can describe a plausible scratch because the prompt asks about defects, or output a non-empty location even when the query is visually consistent with the normal reference [3, 9, 12, 13, 34].

For industrial anomaly reasoning, the failure is often not limited to the final answer. A response can be unusable even when the selected option is correct: the predicted box may not cover the defect, the rationale may mention a visual cue that is absent, or a normal sample may receive a hallucinated defect location. We refer to this as evidence inconsistency among answer, location, and rationale [3, 12, 13, 34]. Such inconsistency is problematic for deployment because industrial users require inspectable decision traces, not merely a multiple-choice answer. In particular, false-positive boxes on normal samples increase manual verification cost and weaken trust in the system.

A straightforward approach is to fine-tune an MLLM on anomaly QA data, but supervised fine-tuning alone mainly encourages imitation of target responses. It does not directly discourage contradictory traces such as a normal answer paired with abnormal evidence, or a correct answer paired with an invalid box. Preference or reinforcement learning can optimize such structured properties [3, 6, 13, 25, 28], but applying it directly to a base MLLM is unstable: the model may not follow the required output format, spatial rewards become ambiguous for normal or non-spatial questions, and answer-only rewards may reinforce shortcut decisions. These observations suggest that anomaly reasoning should be aligned progressively: first establish visual evidence grounding, then introduce domain-oriented instructions, and finally optimize explicit consistency rewards.

We propose **EGVLR**, an Evidence-Grounded Vision-Language Reinforcement framework for industrial anomaly reasoning. EGVLR is organized around a unified Evidence-Driven Diagnostic Protocol (EDDP). Every trainable stage produces the same structured output containing `<evidence>`, `<logic>`, `<location>`,

and `<answer>`. The location field is always a list of normalized bounding boxes, and defect-free, negative-region, or non-spatial samples use `<location>[]</location>`. This simple constraint removes ambiguity between grid labels, boxes, and normal cases, allowing the same parser and reward rules to be used throughout training and rendering.

EGVLR has four stages. **Progressive Visual-Evidential Fine-Tuning (PVE-FT)** first trains visual evidence grounding with synthetic localized anomalies, 3×3 grid supervision, local verification decoys, and null-hypothesis calibration. **Knowledge-Grounded Instruction Tuning (KG-IT)** then teaches industrial-domain QA behavior using domain questions, one-normal visual QA, and comparative inspection QA while preserving the EDDP schema. **Geometry-Semantic Decoupled Preference Optimization (GS-DPO)** further aligns the model by separately rewarding format validity, answer correctness, box geometry, sentence-embedding-based rationale semantics, and answer–location–rationale coupling. Finally, **Box-Guided Segmentation Rendering (BGSR)** maps predicted boxes to dense masks using an external segmentation backend [2, 26]; it is a rendering step and does not change the MLLM answer.

We evaluate EGVLR on MMAD under the industrial anomaly QA setting. The results show that the proposed staged alignment improves evidence-grounded reasoning and reduces spatial false positives compared with general MLLMs and anomaly-oriented baselines [3, 5, 11–14, 34]. Ablations demonstrate that visual pre-alignment, knowledge-grounded tuning, geometry-semantic preference optimization, and box-guided segmentation rendering address different failure modes and are complementary.

Contributions. The main contributions are summarized as follows:

- We introduce a unified Evidence-Driven Diagnostic Protocol (EDDP) that connects visual evidence, diagnostic logic, normalized bounding-box localization, and final answers across all trainable stages.
- We propose Progressive Visual-Evidential Fine-Tuning (PVE-FT), a vision-only pre-alignment stage that teaches localized evidence grounding through synthetic anomalies, grid localization, local decoy verification, and null-hypothesis calibration.
- We design Geometry-Semantic Decoupled Preference Optimization (GS-DPO), which combines answer reward, box reward, sentence-embedding-based rationale semantic reward, and semantic coupling reward to reduce inconsistent reasoning and spatial reward hacking.
- We conduct MMAD-focused experiments and ablations that analyze stage contributions, reward components, normal false-positive behavior, and box-guided segmentation rendering.

2 Related Work

2.1 Industrial Anomaly Detection and Foundation Models

Industrial anomaly detection (IAD) has been extensively studied as image-level anomaly classification and pixel-level anomaly localization. Representative benchmarks such as MVTec AD, VisA, and Real-IAD [1, 30, 35] have supported methods based on reconstruction, feature modeling, patch memory, and discriminative segmentation [27, 33]. Within this line, multi-scale patch representation learning [29], multi-synthesis segmentation training [23], scalar-quantized representations [21], lookup-free and CLIP-guided quantized autoencoding [19, 20], lightweight multi-class adaptation [22], and component-aware logical anomaly segmentation [8] further improve unsupervised, few-shot, and scalable anomaly detection. Recent work also explores instance-aware zero-shot segmentation and prompt-guided anomaly segmentation across different domains [7, 17]. These methods improve transferability and localization, but their outputs are usually anomaly scores, heatmaps, boxes, or masks. They are not designed to answer natural-language questions or provide explicit diagnostic rationales.

EGVLR targets a complementary problem. Instead of replacing image-only anomaly detectors, it studies how an MLLM can produce an inspectable reasoning trace: what evidence is observed, how the evidence supports the decision, where the abnormal region is, and what final answer should be selected. This requires both visual grounding and language-level consistency, which are not directly optimized in most conventional IAD frameworks.

2.2 Multimodal Anomaly Data and Benchmarks

The rise of MLLMs has motivated multimodal anomaly benchmarks that evaluate language-conditioned inspection. MMAD [9] formulates industrial anomaly understanding as multiple-choice QA over object recognition, defect discrimination, localization, description, and analysis. Anomaly instruction datasets such as Anomaly-Instruct-125K [32] further indicate that anomaly-specific post-training data is important for adapting MLLMs to fine-grained industrial evidence. These resources show that general MLLMs are not automatically reliable anomaly reasoners: they often miss small defects, misinterpret benign variations, or localize imprecisely.

Our work follows this line but focuses on the reliability of the reasoning trace rather than only the availability of multimodal anomaly data. We use MMAD as the main evaluation benchmark and design training stages that explicitly address evidence grounding, normal-case calibration, and answer–location–rationale consistency. In this sense, EGVLR complements existing datasets by providing a post-training and reward-design protocol for structured anomaly reasoning.

2.3 MLLMs for Industrial Anomaly Understanding

Several recent methods adapt MLLMs to industrial inspection through prompting, instruction tuning, or task-specific visual modules. AnomalyGPT [5], Myr-

iad [14], FabGPT [10], VMAD [11], Anomaly-OV [32], and WaferMapVQA [16] demonstrate that language supervision can improve anomaly recognition and explanation. Related anomaly-understanding studies also extend vision-language reasoning to medical anomaly segmentation and video anomaly understanding [4, 17]. Other works, including OmniAD [34], JUDO [12], AnomalyR1 [3], and IAD-R1 [13], further explore reference-guided reasoning, domain knowledge, and reinforcement-style training for industrial anomaly QA.

These studies establish that anomaly-oriented post-training is necessary, but several reliability questions remain open. First, a response format may contain a rationale without forcing the rationale to match the final decision. Second, spatial predictions may be evaluated only on abnormal samples, leaving normal-case box hallucination under-specified. Third, domain knowledge and visual grounding are often learned together, making it difficult to tell whether gains come from better visual evidence localization or from stronger textual priors. EGVLR separates these roles: PVE-FT is a vision-only pre-alignment stage, KG-IT injects domain-oriented QA behavior, and GS-DPO explicitly optimizes the consistency among answer, location, and rationale.

2.4 Instruction Tuning and Preference Optimization

Instruction tuning is widely used to adapt MLLMs to downstream tasks [24, 31]. For anomaly reasoning, it can teach prompt following, option selection, domain terminology, and structured response style. However, because SFT is token-level imitation, it does not directly reward the global validity of a generated trace. A model may learn to output the correct tags but still produce a box inconsistent with the answer or a rationale inconsistent with the visual evidence.

Preference optimization and reinforcement-style alignment offer a way to align generated responses with task-level rewards. Recent rule-based reasoning methods and GRPO-style training [6, 28] reduce the need for manually annotated preference pairs by computing rewards from verifiable output properties. For anomaly reasoning, reward design must handle both language and geometry. Answer-only rewards can encourage shortcut prediction, whereas geometry-only rewards can be exploited through oversized or redundant boxes. EGVLR therefore decomposes GS-DPO into format, answer, box, semantic rationale, and coupling channels. The semantic channel uses sentence embeddings computed with the sentence-transformers package to compare generated rationales with reference sentences that represent normal-supporting, abnormal-supporting, and domain-oriented reasoning patterns, while the coupling channel checks whether the predicted answer, location, and rationale direction agree.

Reward-based preference optimization for structured vision-language outputs. Generic preference optimization usually ranks responses according to overall helpfulness or answer correctness. Industrial anomaly reasoning is different: a response is useful only when its answer, spatial evidence, and rationale are mutually consistent. A reward that only scores the final answer may reinforce shortcut decisions, while a reward that only scores boxes may encourage redundant

proposals. EGVLRL therefore uses a geometry–semantic reward decomposition in which answer correctness, IoU-based box matching, sentence-embedding-based semantic similarity, and answer–location–rationale coupling are optimized jointly under the same EDDP parser. This makes the preference objective specific to evidence-grounded industrial inspection rather than generic response preference learning.

2.5 Reference-Guided Reasoning and Box-Prompted Rendering

Industrial defects are often defined relative to normal appearance, so reference-guided comparison is an important cue for anomaly reasoning. A normal reference helps distinguish true defect evidence from benign variation in pose, lighting, texture, or product layout. EGVLRL uses a fixed paired-image convention: the first image is the query/test image and the second image is the normal reference. This convention is used consistently in PVE-FT, KG-IT, and GS-DPO to reduce ambiguity in comparative reasoning.

For dense visualization, segmentation foundation models such as SAM variants [2, 26] can convert spatial prompts into masks. EGVLRL treats this as a rendering problem rather than a reasoning problem. The MLLM predicts normalized boxes under the EDDP schema; BGSR maps non-empty boxes to pixel coordinates and sends them to an off-the-shelf segmentation backend. If the location is empty, no mask is produced. This separation avoids attributing the segmentation backend’s mask quality to the MLLM’s reasoning ability and keeps the training-time contribution distinct from the visualization module.

3 Method

3.1 Overview and Unified EDDP Schema

EGVLRL follows a four-stage pipeline for industrial anomaly reasoning: PVE-FT, KG-IT, GS-DPO, and BGSR. The central design is a unified Evidence-Driven Diagnostic Protocol (EDDP) used by all trainable stages:

$$y = \langle \text{think} \rangle \langle \text{evidence} \rangle e \langle / \text{evidence} \rangle \langle \text{logic} \rangle l \langle / \text{logic} \rangle \langle / \text{think} \rangle \langle \text{location} \rangle B \langle / \text{location} \rangle \langle \text{answer} \rangle a \langle / \text{answer} \rangle. \quad (1)$$

Here, e is the visual or domain evidence, l is the diagnostic logic, a is the final answer, and B is a list of normalized bounding boxes. Each box is represented as $[x_1, y_1, x_2, y_2]$ in the $[0, 1000]$ coordinate system. For normal, negative-region, or non-spatial QA samples, the correct location is the empty list, implemented as $\langle \text{location} \rangle [] \langle / \text{location} \rangle$. The $\langle \text{location} \rangle$ field never stores a grid ID; grid IDs are used only as textual spatial evidence and metadata.

For paired inputs, the image order is fixed throughout the whole framework:

$$\text{first image} = \text{query/test image}, \quad \text{second image} = \text{normal reference image}. \quad (2)$$

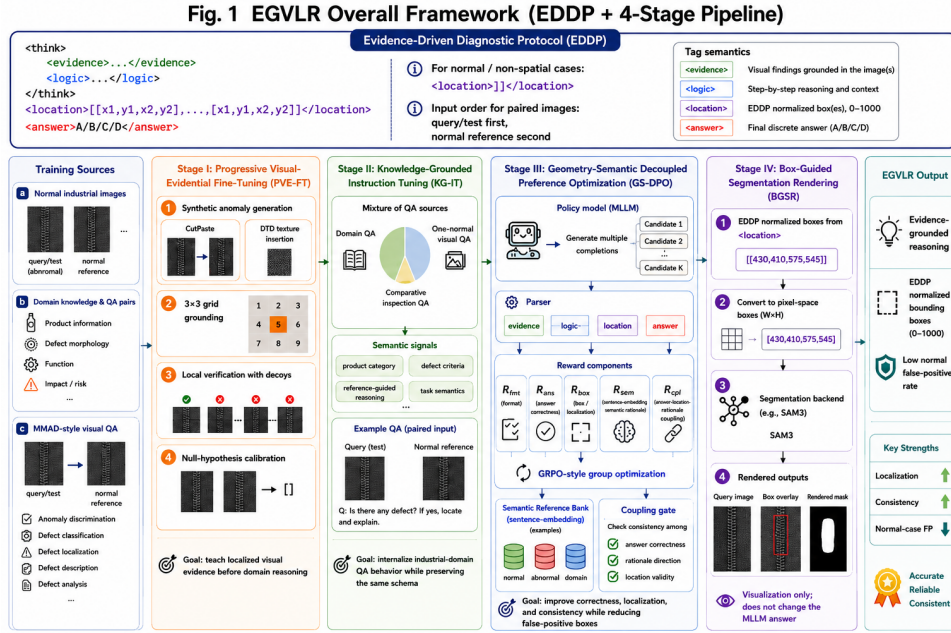


Fig. 1: Overview of EGVLR. The framework uses one unified Evidence-Driven Diagnostic Protocol (EDDP) across all trainable stages. PVE-FT teaches localized visual evidence with synthetic anomalies, grid grounding, local decoys, and null-hypothesis calibration. KG-IT injects domain-oriented and reference-guided QA behavior. GS-DPO aligns answer correctness, box validity, rationale semantics, and answer–location–rationale coupling. BGSR converts non-empty predicted boxes into rendered masks using an external segmentation backend; it does not change the MLLM answer.

This convention removes ambiguity between comparative training, reward parsing, and mask rendering. Figure 1 summarizes the complete pipeline and highlights how the same EDDP schema connects visual pre-alignment, knowledge-grounded tuning, preference optimization, and box-guided rendering.

3.2 Stage I: Progressive Visual-Evidential Fine-Tuning

PVE-FT teaches the MLLM to ground its decision in localized visual evidence before introducing domain-level reasoning. Given a normal image I , we synthesize a query image $\tilde{I}$ by inserting a localized anomaly with CutPaste-style patch replacement or DTD texture insertion. The generator records the anomaly mask M , normalized box B^* , 3×3 grid ID g^* , and optional subregion label s^* . The target rationale is then generated programmatically from this controlled metadata. It only verbalizes observable visual evidence and does not introduce external domain knowledge.

PVE-FT contains five visual-only QA families: synthetic anomaly detection, 3×3 grid localization, fine subregion localization, local verification with pos-

itive and decoy regions, and normal-reference null-hypothesis calibration. For all grid-based questions, only the query image is divided into a 3×3 grid over the displayed image canvas. The full grid convention and prompt templates are provided in the supplementary material.

For abnormal synthetic samples, the target `<location>` field contains the generated box. For normal samples and decoy-region samples, the `<location>` field is empty, i.e., `[]`. PVE-FT is optimized with autoregressive cross-entropy:

$$\mathcal{L}_{\text{PVE}}(\theta) = -\mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{PVE}}} \sum_{t=1}^{|y|} \log \pi_{\theta}(y_t \mid x, y_{<t}). \quad (3)$$

3.3 Stage II: Knowledge-Grounded Instruction Tuning

After visual-evidential pre-alignment, KG-IT teaches the model task semantics, object-level inspection knowledge, and reference-guided anomaly QA behavior. While PVE-FT focuses on localized visual evidence, KG-IT shifts the model toward industrial inspection reasoning under the same EDDP schema. In our MMAD instantiation, KG-IT uses three complementary sources: domain QA generated from domain snippets, one-normal visual QA, and comparative inspection QA. Domain QA introduces object categories, defect terminology, and functional risk cues; one-normal visual QA teaches standard query-reference anomaly reasoning; and comparative inspection QA provides hard-negative calibration by requiring the model to compare subtle deviations against a normal reference. These sources are mixed to balance domain knowledge, visual anomaly QA, and comparative reasoning.

KG-IT preserves the same EDDP schema across all samples. For samples without reliable box supervision, we still keep the `<location>` field and set it to `[]`, while marking them as `require_location=false` for later preference optimization. This design avoids format drift between PVE-FT, KG-IT, and GS-DPO: the model always learns to produce evidence, logic, location, and answer fields, but only samples with reliable spatial supervision contribute to the spatial reward later. KG-IT is optimized with the standard autoregressive supervised objective:

$$\mathcal{L}_{\text{KG}}(\theta) = -\mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{KG}}} \sum_{t=1}^{|y|} \log \pi_{\theta}(y_t \mid x, y_{<t}). \quad (4)$$

This stage provides a stable domain-aware initialization for GS-DPO, so that preference optimization starts from a model that already follows the EDDP format and understands the basic semantics of industrial anomaly QA.

3.4 Stage III: Geometry-Semantic Decoupled Preference Optimization

Instruction tuning teaches the model to follow the EDDP schema, but token-level imitation alone cannot explicitly penalize globally inconsistent outputs. A

completion may contain valid tags and even a correct answer, while still producing an invalid box or a rationale that contradicts the visual evidence. GS-DPO therefore refines the KG-IT model with a GRPO-style objective and decouples the reward into format, answer, geometry, semantic rationale, and coupling components.

For each prompt q , the policy samples a group of G completions $\{o_i\}_{i=1}^G$ and assigns a reward R_i to each completion. The group-normalized advantage is:

$$\hat{A}_i = \frac{R_i - \mu_R}{\sigma_R + \epsilon}, \quad \mu_R = \frac{1}{G} \sum_{i=1}^G R_i, \quad \sigma_R = \sqrt{\frac{1}{G} \sum_{i=1}^G (R_i - \mu_R)^2}. \quad (5)$$

When the group reward variance is zero, we set the normalized advantages to zero to avoid unstable updates. The policy objective is:

$$\mathcal{J}_{\text{GS}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \min \left(\rho_i(\theta) \hat{A}_i, \text{clip}(\rho_i(\theta), 1 - \epsilon_c, 1 + \epsilon_c) \hat{A}_i \right) - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \right], \quad (6)$$

where $\rho_i(\theta) = \pi_\theta(o_i | q) / \pi_{\theta_{\text{old}}}(o_i | q)$.

Decoupled reward. The total reward is:

$$R = \lambda_f R_{\text{fmt}} + \lambda_a R_{\text{ans}} + \lambda_b R_{\text{box}} + \lambda_r R_{\text{sem}} + \lambda_c R_{\text{cpl}}. \quad (7)$$

Here, R_{fmt} checks whether the EDDP fields are parseable, and $R_{\text{ans}} = \mathbb{I}(a_{\text{pred}} = a_{\text{gt}})$ rewards answer correctness. For abnormal samples with spatial supervision, R_{box} matches predicted and ground-truth boxes using IoU-based matching, with a count penalty for redundant boxes. For normal or negative-region samples, $R_{\text{box}} = 1$ if the predicted location is empty and -1 otherwise. For non-spatial QA marked as `require_location=false`, $R_{\text{box}} = 0$.

For rationale semantics, we concatenate the generated `<evidence>` and `<logic>` fields and encode the resulting rationale using a lightweight sentence embedding model. Let $\mathbf{z}$ denote the normalized embedding of the generated rationale. We maintain three fixed semantic reference banks: a normal-supporting bank $\mathcal{B}_N$, an abnormal-supporting bank $\mathcal{B}_A$, and a domain-oriented bank $\mathcal{B}_D$. The similarity to each bank is computed by maximum cosine similarity:

$$s_N = \max_{\mathbf{u} \in \mathcal{B}_N} \cos(\mathbf{z}, \mathbf{u}), \quad s_A = \max_{\mathbf{u} \in \mathcal{B}_A} \cos(\mathbf{z}, \mathbf{u}), \quad s_D = \max_{\mathbf{u} \in \mathcal{B}_D} \cos(\mathbf{z}, \mathbf{u}). \quad (8)$$

The semantic rationale reward encourages the rationale to follow the expected semantic direction:

$$R_{\text{sem}} = \begin{cases} \tanh(\gamma(s_A - s_N)), & \text{abnormal visual samples,} \\ \tanh(\gamma(s_N - s_A)), & \text{normal or negative-region samples,} \\ \tanh(\gamma(s_D - \max(s_N, s_A))), & \text{domain-only QA.} \end{cases} \quad (9)$$

Thus, abnormal samples are rewarded when their rationales are closer to abnormal-supporting references, normal or negative-region samples are rewarded when their rationales are closer to normal-supporting references, and domain-only QA is rewarded when its rationale is closer to domain-oriented references. The parameter γ controls reward sharpness, and $\tanh(\cdot)$ bounds the reward to a stable range.

The coupling reward further enforces consistency among the final answer, predicted location, and rationale direction:

$$R_{\text{cpl}} = \mathbb{I}(a_{\text{pred}} = a_{\text{gt}}) \cdot \mathbb{I}(\text{LocValid}) \cdot \mathbb{I}(\text{SemValid}). \quad (10)$$

Here, `LocValid` checks whether the predicted location is valid for the current sample type: normal and negative-region samples require an empty location, whereas abnormal samples require a valid matched box. For non-spatial QA, location validity is treated as satisfied because no spatial evidence is required. `SemValid` checks whether the rationale direction agrees with the sample type using the same semantic margins as R_{sem} . Therefore, R_{cpl} is positive only when the answer is correct, the location is valid, and the rationale is semantically consistent. This design penalizes redundant boxes, non-empty boxes on normal images, answer-correct but location-invalid completions, and rationales that contradict the final answer.

Malformed outputs. All completions are parsed by the same deterministic EDDP parser during training and evaluation. If a completion misses required fields, contains an invalid answer token, or produces malformed locations, it receives the minimum format reward and its invalid location is treated as non-matching for spatial reward computation. For normal, negative-region, or non-spatial samples, any non-empty malformed or unparseable location is counted as a false-positive location. This rule avoids manually repairing invalid outputs and keeps reward computation identical across ablations.

3.5 Stage IV: Box-Guided Segmentation Rendering

BGSR does not further train or modify the MLLM. It only converts the predicted `<location>` field into dense visual masks. If the predicted location is empty, no mask is generated. Otherwise, each normalized box is mapped back to the original image resolution and used as a box prompt for an off-the-shelf segmentation backend. In our default implementation, we use a SAM-family segmentation backend for rendering. This stage decouples cognitive anomaly reasoning from dense mask rendering: the MLLM is responsible for evidence-grounded reasoning and box localization, while the segmentation backend only renders pixel-level masks from verified boxes.

4 Experiments

4.1 Experimental Setup

Dataset. We evaluate on MMAD [9], a multimodal benchmark for industrial anomaly understanding. MMAD evaluates anomaly discrimination and reasoning-dependent defect QA, including defect classification, localization, description, and analysis. All medical-domain experiments are omitted from the main paper to keep the evaluation focused on industrial anomaly reasoning.

Backbone and training. We instantiate EGVLR with Qwen3-VL-8B-Instruct as the backbone.

For paired-image inputs, the first image is the query/test image and the second image is a normal reference image from the same object category.

Following recent industrial anomaly reasoning practice, PVE-FT uses synthetic visual-evidential QA generated from normal industrial images and a small amount of Real-IAD visual data for Stage-I visual pre-alignment [12, 30]. When any MMAD images are sampled for auxiliary training, the corresponding samples are excluded from MMAD evaluation to avoid data leakage [12]. KG-IT uses the mixed instruction data described in Sec. 3.3. GS-DPO is initialized from the KG-IT checkpoint and optimized with the reward in Eq. 7. BGSr is applied only for visualization and does not modify model answers.

Implementation details. We use a fixed EDDP parser for all stages and evaluate only the parsed final answer and parsed location field. All predicted boxes are represented in the query-image coordinate frame using EDDP normalized 0–1000 coordinates and are converted back to pixel coordinates only for evaluation or BGSr rendering. Reward coefficients and validity thresholds are used as fixed default settings across the reported ablations rather than being tuned separately for each setting. Malformed outputs are not manually repaired; they are handled by the deterministic parser and reward rules described in Sec. 3.4. Further implementation details are provided in the supplementary material.

Baselines and metrics. We compare against general-purpose MLLMs, open-source VLMs, and anomaly-specialized baselines, including Qwen-VL variants, LLaVA-style models, AnomalyGPT, OmniAD, AnomalyR1, IAD-R1, and JUDO when available. For MMAD QA tasks, we report accuracy. To evaluate whether a model produces reliable spatial evidence, we additionally report localization accuracy, EDDP format validity, and two normal-case spatial false-positive metrics. Format measures the percentage of outputs that pass the deterministic EDDP parser, including required tags, a valid answer token, and a parseable `<location>` field. FPR_{img} measures the percentage of normal samples for which the model predicts a non-empty `<location>` field:

$$\text{FPR}_{img} = \frac{|\{x \in \mathcal{N} : \hat{B}(x) \neq \emptyset\}|}{|\mathcal{N}|}, \quad \text{FPBox} = \frac{1}{|\mathcal{N}|} \sum_{x \in \mathcal{N}} |\hat{B}(x)|. \quad (11)$$

Table 1: Representative comparison on the MMAD industrial benchmark. We report seven-task accuracy following the MMAD evaluation protocol when available. Results from methods with different inference mechanisms should be interpreted with protocol differences in mind. The best results in this representative comparison are highlighted in **bold** and the second-best are underlined.

Model		Anomaly Scale	Defect Discrim.	Defect Class.	Defect Local.	Defect Desc.	Defect Analysis	Object Class.	Object Analysis	Object Average
<i>Commercial Models</i>										
GPT-4o	-	68.63	65.80	55.62	73.21	83.41	<u>94.98</u>	82.80		74.92
Gemini 2.5 Pro	-	83.07	73.86	67.20	79.97	86.27	94.88	83.08		<u>81.19</u>
<i>Open-Source & Specialized</i>										
Qwen2.5-VL-Instruct	7B	<u>71.39</u>	54.35	61.17	65.81	79.32	91.44	84.43		72.56
LLaVA-OneVision-1.5	8B	60.85	56.29	55.25	74.73	83.21	91.03	89.14		72.93
Qwen3-VL-Instruct (Base)	8B	69.75	59.34	59.08	73.54	81.29	90.30	<u>89.07</u>		74.62
AnomalyR1	7B	60.93	64.81	70.72	79.06	85.52	93.12	86.91		77.29
OmniAD	7B	68.80	78.80	75.50	67.20	86.40	96.00	86.40		79.90
JUDO	7B	64.51	72.17	<u>75.95</u>	<u>84.38</u>	<u>87.76</u>	94.24	86.07		80.73
EGVLR (Ours)	8B	66.66	<u>74.57</u>	75.99	84.98	88.37	90.74	88.16		81.35

Here, $\mathcal{N}$ denotes the set of normal samples and $\hat{B}(x)$ denotes the predicted box list for sample x . A lower FPR_{img} indicates fewer hallucinated defect boxes on normal images, while FPBox measures the average number of predicted boxes on normal samples. These metrics complement MMAD QA accuracy because an answer can be correct while its predicted spatial evidence is still invalid.

4.2 Main Results on MMAD

Table 1 reports the main comparison on MMAD. EGVLR achieves highly competitive performance among recent MMAD-oriented industrial anomaly reasoning methods, with particularly strong results on defect-centered localization, description, and analysis. The gains are most visible when the task requires the model to connect localized evidence with a final answer. Compared with anomaly-specialized baselines in this representative table, EGVLR improves spatial reliability by reducing non-empty boxes on normal samples.

Backbone effect. We also compare our Qwen2.5-VL-7B and Qwen3-VL-8B EGVLR variants to separate backbone scaling from evidence-grounded alignment. Reported open-source MMAD one-shot baselines show that Qwen3-VL-8B improves the average score over Qwen2.5-VL-7B, but its native defect localization score is lower. In contrast, after applying EGVLR, the Qwen3-based variant slightly surpasses the Qwen2.5-based variant in defect localization. This suggests that PVE-FT and GS-DPO compensate for Qwen3’s native localization weakness by explicitly aligning localized evidence, normalized boxes, and rationale semantics; the spatial gains are therefore mainly attributed to the proposed evidence-grounded alignment pipeline rather than backbone scaling alone.

Table 2: Stage ablation on MMAD using the Qwen3-VL-8B backbone. Avg. Acc. is the seven-task MMAD average, while Defect Loc. Acc. is the MMAD defect-localization task accuracy. We additionally report EDDP format validity and normal-case spatial false positives. FPR_{img} is the percentage of normal samples with a non-empty predicted `<location>` field, and FPBox is the average number of predicted boxes per normal sample, including samples with zero predicted boxes. BGSr is omitted because it only renders masks from predicted boxes and does not modify the MLLM answer or box predictions.

Variant	Avg. Acc. (%)	Defect Loc. Acc. (%)	Format (%)	$FPR_{img} \downarrow$ (%)	FPBox $\downarrow$
Base MLLM	74.62	59.08	58.7	42.6	0.98
+ PVE-FT	77.84	71.90	94.6	29.4	0.67
+ PVE-FT + KG-IT	80.63	73.42	97.9	21.8	0.50
+ PVE-FT + KG-IT + GS-DPO	81.35	75.99	99.6	7.9	0.18

Table 3: Reward ablation for GS-DPO on MMAD using the Qwen3-VL-8B backbone and the same KG-IT initialization. Answer-only GRPO uses only R_{ans} , while standard GRPO uses the deterministic format, answer, and box rewards without the proposed semantic rationale and coupling rewards. The full objective combines R_{fmt} , R_{ans} , R_{box} , R_{sem} , and R_{cpl} . Lower FPR_{img} and FPBox indicate fewer hallucinated boxes on normal samples.

Variant	Avg. Acc. (%)	Defect Loc. Acc. (%)	Format (%)	$FPR_{img} \downarrow$ (%)	FPBox $\downarrow$
Answer-only GRPO (R_{ans})	80.12	72.84	95.4	18.6	0.36
Standard GRPO ($R_{fmt}+R_{ans}+R_{box}$)	80.74	75.12	98.8	12.8	0.25
w/o box reward	80.41	71.66	98.9	16.9	0.33
w/o semantic rationale reward	80.88	74.83	99.2	10.3	0.22
w/o semantic coupling reward	81.02	75.42	99.3	13.7	0.27
Full GS-DPO	81.35	75.99	99.6	7.9	0.18

4.3 Stage and Reward Ablation

We ablate the contribution of each stage and reward component in Tables 2 and 3. PVE-FT improves format validity and coarse localization because it teaches the EDDP schema and visual grounding before domain QA. KG-IT improves task-level reasoning by injecting domain-oriented instruction behavior. GS-DPO further improves consistency-sensitive metrics, especially localization and normal-case false-positive control.

Beyond QA accuracy, FPR_{img} and FPBox capture normal-case spatial hallucination: a model may answer a normal case correctly while still producing a spurious non-empty `<location>` field. Their reduction shows that the proposed staged alignment and GS-DPO improve spatial reliability, not merely option-selection accuracy.

In reward ablation, answer-only optimization does not improve final-choice accuracy over the KG-IT initialization and is insufficient for reliable anomaly reasoning. Removing the box reward weakens localization; removing the semantic rationale reward weakens semantic alignment; removing the coupling reward increases answer–location–rationale contradictions and normal-case spatial false positives.

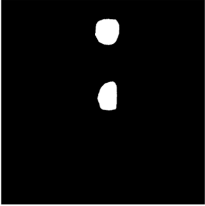
	Normal Image	Query Image	G.T
			
Model	Diagnostic reasoning trace		Prediction
LLaVA-OneVision (8B)	The object is identified as a zipper, but the response attributes the defect to missing teeth and describes non-existent gaps along the track. This is a language-prior hallucination: the explanation is plausible for a zipper, but it is not supported by the visual evidence.		× Missing teeth
Qwen3-VL-Instruct (8B)	The response predicts a defect but gives a generic description about irregular coloration and border structure. It fails to isolate the true morphological cue on the zipper teeth and does not provide a reliable localized decision trace.		× Broken border
EGVLR (Ours) (8B)	<code><think></code> <code><evidence></code> Compared with the normal reference, the fabric border remains intact, while two localized regions on the center zipper track show teeth pushed closer together without clear missing gaps. <code></evidence></code> <code><logic></code> The abnormal cue is a spacing deformation of the teeth rather than missing parts or border damage; therefore the defect corresponds to squeezed zipper teeth. <code></logic></code> <code></think></code> <code><location>[[430,90,570,230],[450,380,560,540]]</location></code> <code><answer>C</answer></code>		✓ Teeth pushed closer together

Fig. 2: Qualitative comparison on a zipper anomaly example. The visual pair contains a normal reference and a query image whose teeth are pushed closer together. Baselines produce plausible but weakly grounded defect descriptions, whereas EGVLR follows the unified EDDP schema, localizes the multi-instance defect with EDDP normalized 0–1000 boxes, and predicts the correct answer.

4.4 Qualitative Analysis

Figure 2 shows qualitative examples on MMAD. General-purpose MLLMs often produce plausible explanations without reliable spatial grounding. In contrast, EGVLR first identifies localized evidence, then predicts a bounding box, and finally provides the answer. On normal samples, EGVLR tends to retain the null hypothesis and outputs an empty location. This behavior is consistent with the design of PVE-FT and GS-DPO.

4.5 Failure Cases

Although EGVLR improves evidence-grounded anomaly reasoning, several failure cases remain. Extremely small or low-contrast defects may still be missed, especially when the defect resembles normal texture variation or when the query-reference difference is dominated by illumination, pose, or product-layout changes.

When multiple defects appear in one image, the model may localize only the most salient region and miss secondary defects, indicating that stronger multi-instance localization rewards are still needed. Normal-case calibration also remains challenging when benign manufacturing variation visually resembles defect evidence; in such cases, the model may either over-suppress subtle anomalies or produce a weak false-positive location. The sentence-embedding-based semantic reward is a lightweight text-based signal over semantic reference banks and does not by itself verify whether a rationale is visually true. Therefore, broader cross-dataset evaluation, detector-assisted MLLM comparisons, and stronger visual-semantic verification remain important future work rather than claims made by the present MMAD-focused evaluation.

5 Conclusion

We presented EGVLR, an evidence-grounded vision–language reinforcement framework for industrial anomaly reasoning on MMAD. EGVLR uses a unified EDDP schema to connect visual evidence, diagnostic logic, bounding-box localization, and final answers across all trainable stages. PVE-FT teaches the model to localize visual evidence and retain the null hypothesis for normal samples. KG-IT injects domain-oriented QA behavior while preserving the same output schema. GS-DPO further aligns answer correctness, spatial validity, sentence-embedding-based rationale semantics, and answer–location–rationale consistency. Finally, BGSR converts predicted boxes into dense masks using an external segmentation backend. Experiments and ablations on MMAD show that EGVLR improves reasoning-dependent anomaly QA and reduces spatial false positives. Future work includes stronger multi-instance localization rewards, improved contradiction-aware rationale scoring, and interactive inspection workflows.

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